

Identify GSF member and non-member organizations relevant to OpenTelemetry and Sustainability

- Cloud providers: Google, Microsoft
- Observability providers: Sentry, New Relic (not a member, but I have someone in mind)
- OpenTelemetry user: Goldman Sachs, Mastercard, Bloomberg, Amadeus etc

Define the key personas we need to serve, including enterprise OTEL users and OTEL maintainers

- 1. The Enterprise OTel User (The Consumer)
 - This persona represents DevOps engineers, SREs, and Platform Engineers at large organizations who are already using OpenTelemetry for performance monitoring.
 - Their primary goal is to fulfill corporate sustainability mandates (Scope 3 reporting) without adding significant operational overhead or proprietary tooling.
- 2. The OTel Maintainer (The Enabler)
 - This persona includes the contributors and maintainers of the OpenTelemetry project, specifically those working on the Collector and Semantic Conventions.
 - Their goal is to ensure that any new sustainability metrics are performant, vendor-neutral, and adhere to the strict architectural standards of the OpenTelemetry project.
- 3. The Sustainability Lead (The Stakeholder)
 - While not a direct technical user of OTel, this persona relies on the data generated by the system.
 - Primary Goal: To obtain granular, auditable data for ESG (Environmental, Social, and Governance) reporting.

Draft a short PR/FAQ document for an SCI for OpenTelemetry specification guideline

PRFAQ: Software Carbon Intensity (SCI) for OpenTelemetry

Press Release

Standardizing Carbon Observability with SCI for OpenTelemetry

LONDON, UK — The Green Software Foundation (GSF) announces the launch of the "**SCI for OpenTelemetry**" project, an initiative to bridge cloud-native observability and environmental sustainability by providing a standardized, automated way to calculate and export [Software Carbon Intensity \(SCI\)](#) scores directly through [OpenTelemetry](#).

As organizations face increasing pressure to report on Scope 3 emissions, the lack of granular, real-time software-level carbon data remains a significant hurdle. While tools like [Kepler](#) and [Aether](#) have made strides in measuring energy consumption, there is no unified standard for injecting these metrics into OpenTelemetry, the industry-standard observability framework.

"By integrating the SCI directly into the OpenTelemetry protocol, we are moving sustainability from a post-hoc reporting exercise to a real-time engineering metric. Developers will see their carbon footprint alongside their latency and error rates." — GSF Spokesperson

The project builds upon work by [Sentry Software](#) and [RETIT / University of Applied Sciences Munich](#), who demonstrated that auto-instrumentation agents can be extended to publish green metrics. SCI for OpenTelemetry will formalize these patterns into reusable OTel Collector components and semantic conventions for carbon telemetry.

Frequently Asked Questions

1. What is the SCI for OpenTelemetry project?

A technical initiative to create standardized OpenTelemetry components (receivers, processors, exporters) that calculate the SCI score for a software component. The SCI is defined as:

$$SCI=(E \times I)+MR$$

Where: E = Energy consumed, I = Location-based marginal carbon intensity, M = Embodied carbon of hardware, R = Functional unit (e.g., per request, per user).

2. Why OpenTelemetry instead of a standalone carbon tooling?

OpenTelemetry is the [industry standard for observability](#). By using OTel, organizations leverage existing methodologies to collect carbon data without proprietary agents.

3. What are the primary deliverables?

OTel Semantic Conventions	Standard names for carbon metrics (e.g., hw.energy, sw.carbon_intensity)
SCI Component	OTel Collector component that calculates SCI scores in real time

4. How will this help developers reduce carbon?

By surfacing carbon impact in existing Grafana dashboards, developers can perform "**Carbon A/B testing**", e.g., measuring whether a code optimization reduced the SCI score per 1,000 requests, making environmental impact visible and actionable.

5. Comparison of Carbon Data Sources for OTel Integration

Keple	Energy Consumption (EE)	Kubernetes / eBPF	Kepler Project
Aether	Cloud VM Emissions	AWS / Azure / GCP	Aether GitHub
Hardware Sentry	Data Center Power/Cooling	Physical Servers	Sentry Software
Real-Time Cloud	Carbon Intensity (II)	Global Grid Data	GSF Real-Time Cloud